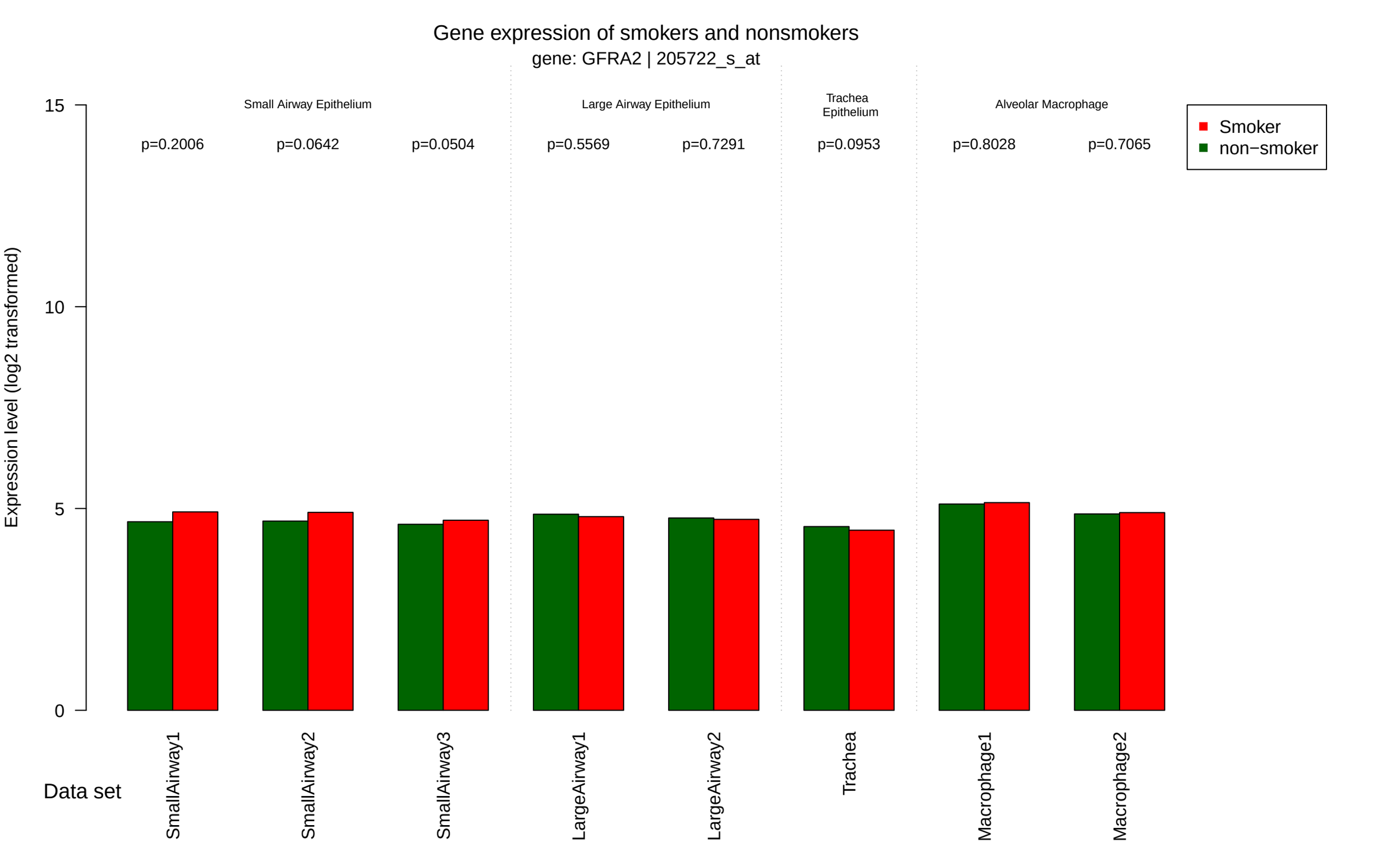
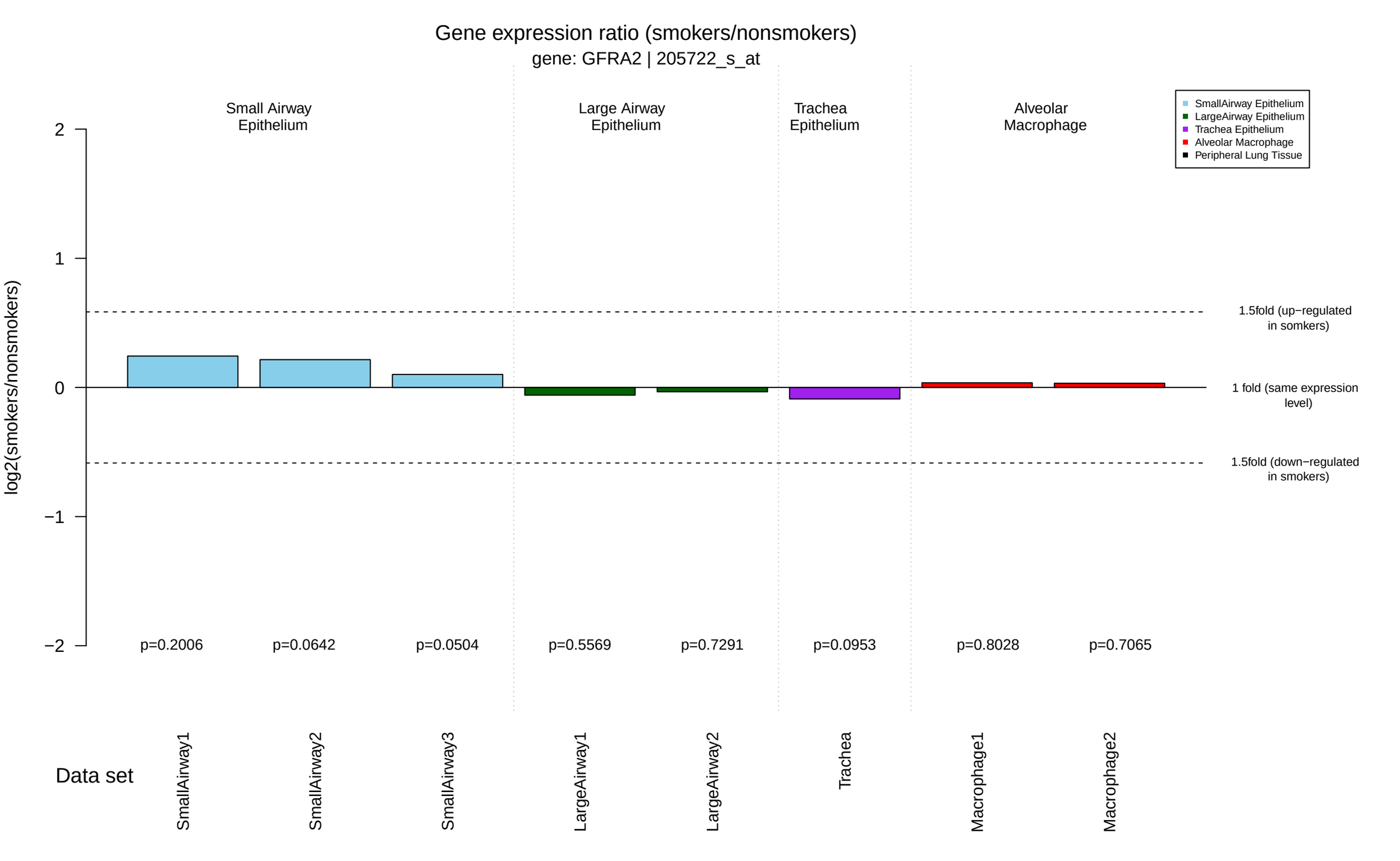
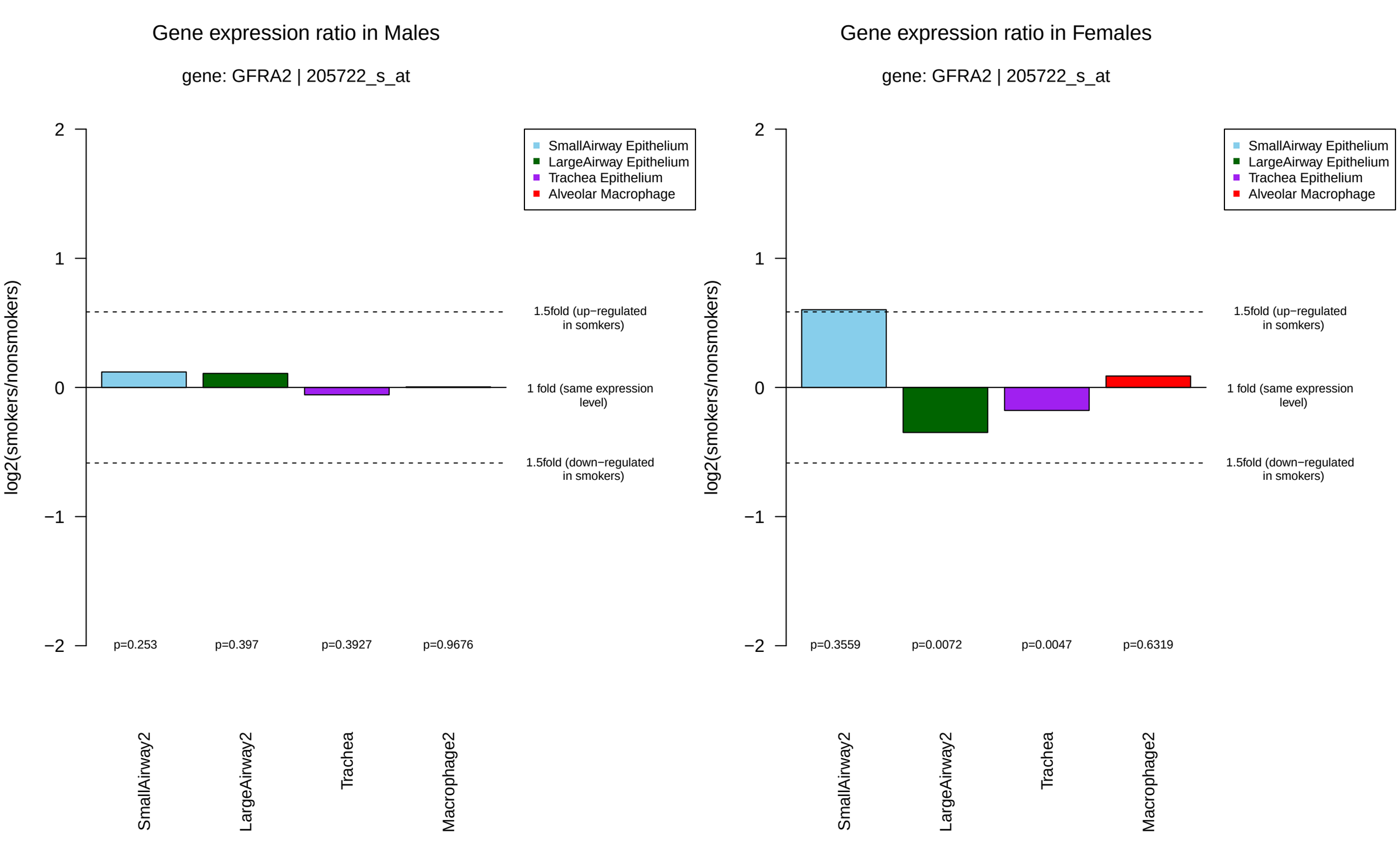
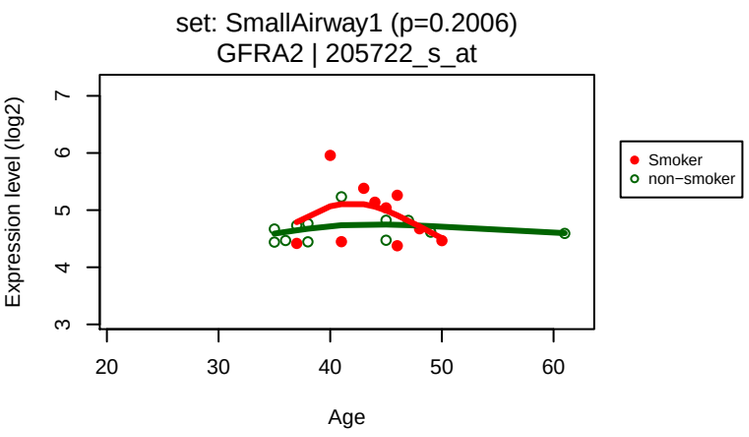
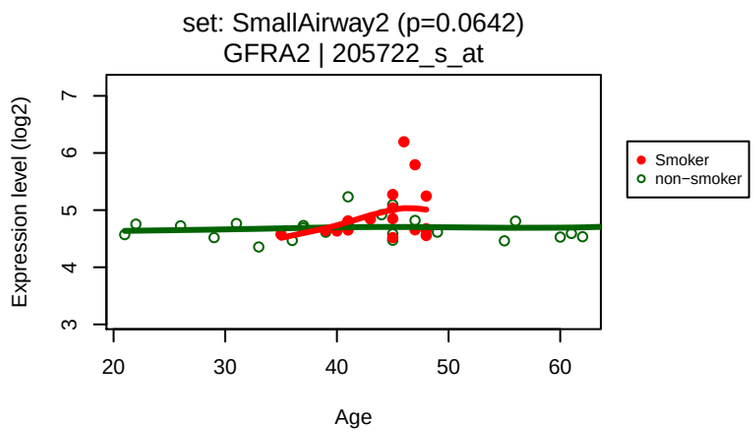
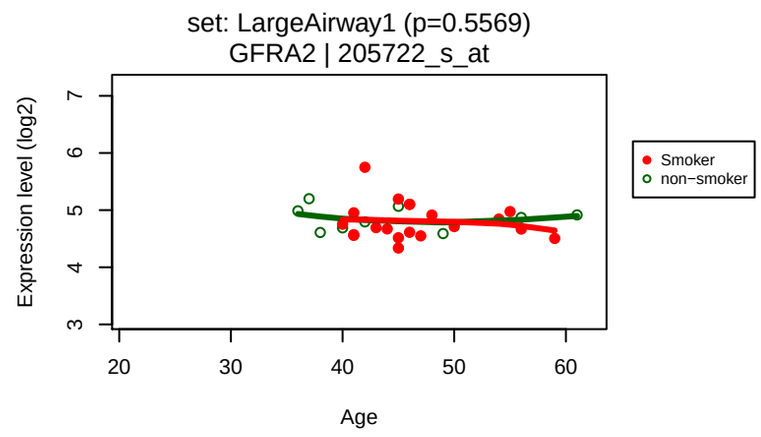

Gene expression of smokers and nonsmokers
gene: GFRA2 | 205722_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2006
p=0.0642
p=0.0504
p=0.5569
p=0.7291
p=0.0953
p=0.8028
p=0.7065
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

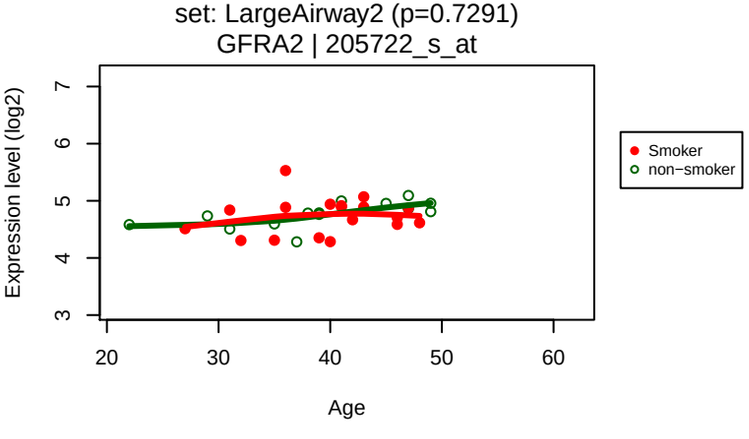

Gene expression ratio (smokers/nonsmokers)
gene: GFRA2 | 205722_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2006
p=0.0642
p=0.0504
p=0.5569
p=0.7291
p=0.0953
p=0.8028
p=0.7065
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

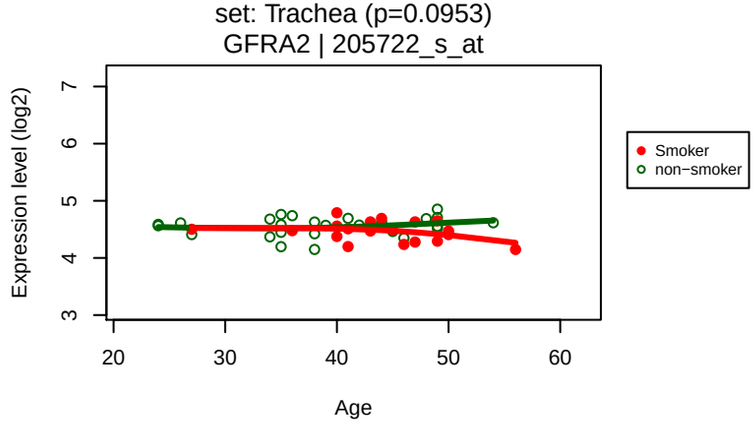

Gene expression ratio in Males
gene: GFRA2 | 205722_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.253
p=0.397
p=0.3927
p=0.9676
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GFRA2 | 205722_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3559
p=0.0072
p=0.0047
p=0.6319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

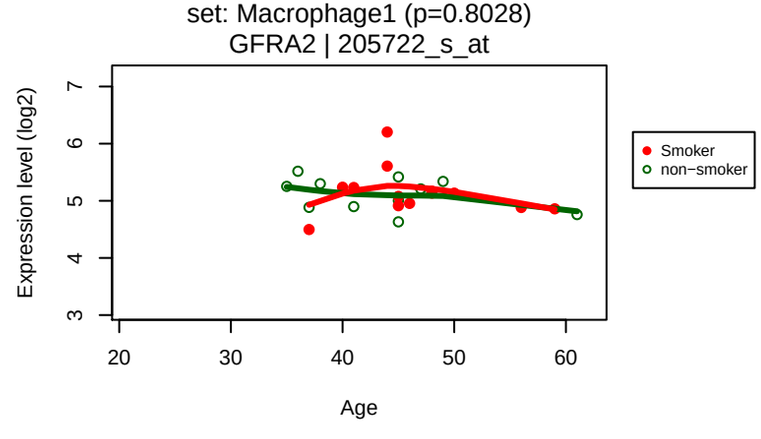

set: SmallAirway1 (p=0.2006)
GFRA2 | 205722_s_at
Expression level (log2)
Age
Smoker
non−smoker

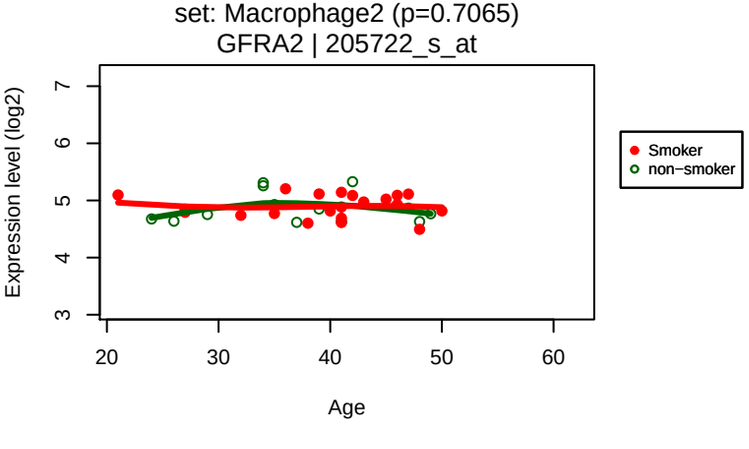

set: SmallAirway2 (p=0.0642)
GFRA2 | 205722_s_at
Expression level (log2)
Age
Smoker
non−smoker

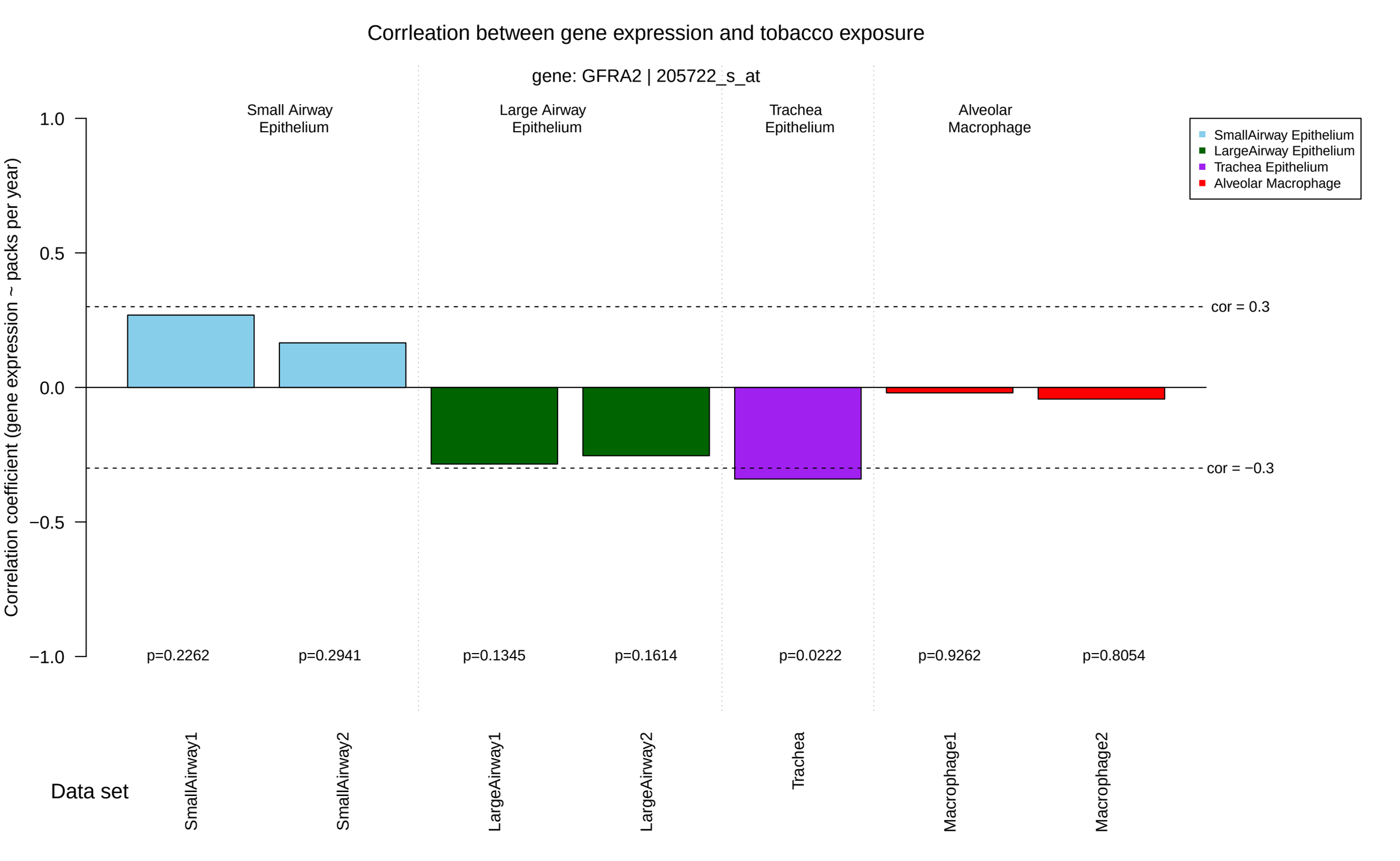

set: LargeAirway1 (p=0.5569)
GFRA2 | 205722_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7291)
GFRA2 | 205722_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0953)
GFRA2 | 205722_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8028)
GFRA2 | 205722_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7065)
GFRA2 | 205722_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GFRA2 | 205722_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2262
p=0.2941
p=0.1345
p=0.1614
p=0.0222
p=0.9262
p=0.8054
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2